

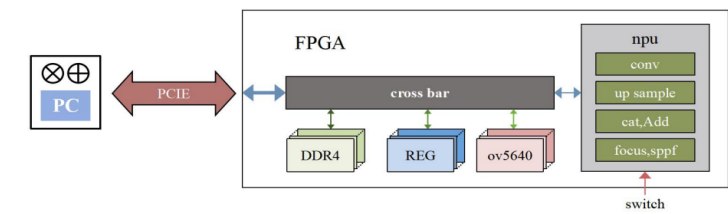
[Public]

Title :FlowRisc

INT8 Stream NPU: Faster AI, Less Power

INTRODUCTION

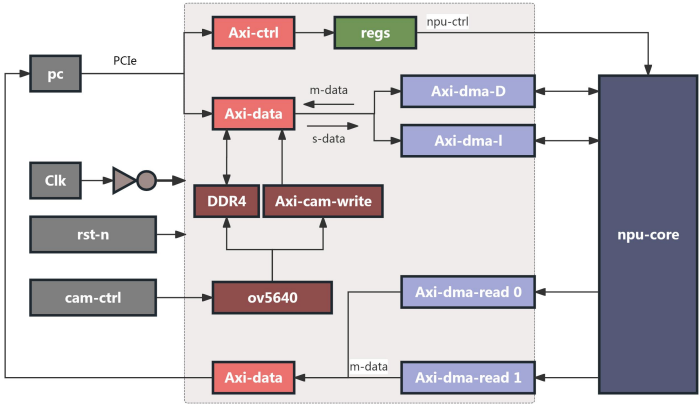
Neural processing units consist of a control plane, a stream_s witch fabric, and composible accelerator cores (Conv3×3, Con v1×1, Add, Concat, upsample, SPPF/Focus) organized into a programmable pipeline.



A typical NPU system consists of a control plane, a stream-switch interconnect, composible accelerator cores , a memory/DMA subsystem, and a host interface. Most current designs rely on vendor IP blocks and opaque subsystems; they support fixed operator acceleration and quantized execution, but often do not expose raw intermediate tensors, fine-grained schedules, or low-level counters, losing transparency, observability, and detailed control.

key technical issue abstract

1. Operator compute, INT8 NPU cores (Conv3×3/1×1, Add, Concat, Upsample, SPPF/Focus) with stream-switch and double-buffered pipelines.
2. Control and scheduling, PCIe XDMA BAR/AXI-Lite register interface (reg_ctrl) with MSI-X status/interrupts.
3. Data movement, 128-bit AXI DMA to DDR4 with crossbar arbitration and ping-pong buffering for overlap of read/compute/ write.
4. I/O ingest and results, dual camera frames to DDR4 and PCIe C2H result streaming with host visualization and logging.



System Architecture

- Programmable-Operator NPU:

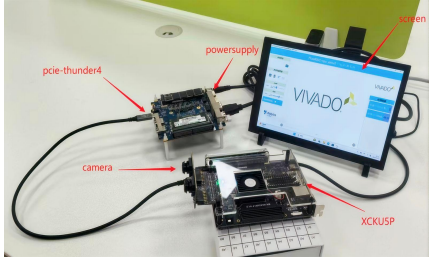
1. Communicating with the host scheduler and registers,fast;
2. Receiving and executing operator graphs ;
3. Acting as the core to speed up inference via overlapped read – compute – write;
4. Collaborating with DMA/DDR and PCIe for data uplink and downlink.

- ProgrammableInferenceOrchestrator:

1. Scheduling and parameterizing operator graphs at runtime;
2. Receiving and staging INT8 tensors from DDR with prefetch and alignment;
3. Streaming results to the host via PCIe for real-time feedback.

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OpenHW2025



On board test by AMD KU5P

CREATIVE DESIGN

RESULT

The completed work has verified the architecture of using an INT8 NPU to accelerate CNN inference with dual camera ingest to DDR and PCIe host interaction. End-to-end tests confirmed the control/data plane

The system is composed of independent components with a flexible and highly scalable architecture. It is based on FPGA logic programmability and high - bandwidth AXI/crossbar/DMA support, which enables it to meet higher performance requirements by adjusting parallelism, operator composition, and dataflow tiling without hardware redesign.

Performance: Throughput bounded by DDR and on-chip memory; 128-bit streams and multi-DMA improve utilization, but memory bandwidth remains the key bottleneck.

Outlook: Add Depthwise/MatMul(+Softmax/LN), command FIFO and operator fusion, and larger on-chip buffering to cut DDR round-trips,Add risc-v process unit to support RV-V order set broaden ,model coverage, and boost energy efficiency.

